RAJADHANI INSTITUTE OF ENGINEERING AND TECHNOLOGY

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Approved by AICTE & Affiliated to APJ Abdul Kalam Technological University
Degree awarded by KTU

KSERC AI Truing-Up Tool

An AI-Driven Innovation for Regulatory Excellence

■ What This Actually Means: What You Pay Per Unit for Electricity

Simple Example: The truing-up process decides your electricity bill amount.

Scenario	What Happens	Effect on Your Bill
Power company estimated costs too high	Actual expenses were less than expected	**Your bill GOES DOWN ↓**
Power company underestimated costs	Actual expenses were more than expected	**Your bill GOES UP ↑**
KSERC finds wasted money or overspending	Some costs are rejected or capped	**Your bill GOES DOWN ↓**
No major differences found	Estimates matched reality	**Bill stays STABLE →**

Real World Example: If Technopark estimated they'd spend ■10 Crore on operations but actually spent ■8 Crore, KSERC checks this during truing-up. If it's legitimate (not wasted), Technopark gets back the difference. If there was wasteful spending, KSERC rejects it. This affects how much per unit Technopark can charge you.

■ What We Built

An intelligent system that automates Kerala's electricity tariff-setting process (truing-up). It combines AI-powered document analysis with KSERC's regulatory framework to process complex financial petitions in **2-4 weeks instead of 6-12 months**.

■ The Challenge We Solve

Manual & Slow	Truing-up took 6-12 months. Officers manually reviewed 500+ page petitions.
Complex Rules	KSERC regulations (12+ rules, multiple exceptions) applied inconsistently.
Data Entry Errors	Thousands of cost items manually copied from PDFs. High error risk.
Lack of Transparency	Limited audit trail. Decisions not fully documented or explainable.

■ How It Works (3 Steps)

1. **Extract:** Upload PDF → AI automatically extracts cost heads, revenue, and financials
2. **Analyze:** AI applies all KSERC regulations & flags deviations with reasoning
3. **Review & Report:** Officers review, override if needed, then generate formal orders

■ Key Features

✓ **Automated PDF Extraction** – No manual data entry
✓ **Smart Prudence Checks** – All regulations applied instantly
✓ **Human Override** – Officers maintain full control
✓ **Auto Report Generation** – Formal KSERC orders (PDF/DOCX)
✓ **Full Audit Trail** – Complete transparency & accountability
✓ **Secure Storage** – Supabase + encryption, local-first option

Impact: 6-12 month process → 2-4 weeks | 100% regulatory compliance | Full transparency for citizens & courts

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Current Implementation & FAQ

Live Deployment & Common Questions

■ Live Deployment Status

Currently processing FY 2024-25 truing-up petitions

Licensee	Type	Status
KDHPCL	Plantation	✓ Active
Technopark	IT Park	✓ Active
Infopark Kerala	IT Park	✓ Active
CSEZA	Special Economic Zone	✓ Active

■ Technology Stack

Frontend	Next.js + React + TypeScript
AI	Claude API + Gemini + Groq fallback
Database	Supabase (PostgreSQL) + local storage
Deployment	Vercel + GitHub + self-hosted option

■ Common Questions & Answers

Can the AI make mistakes?

Yes. That's why it's 'AI-assisted, human-verified.' Every AI decision is flagged for officer review. Officers can override with documented reasoning. Full audit trail maintained.

Is our data safe?

All data encrypted in Supabase. PDFs not permanently stored—only extracted data retained. Full audit logs. Option for on-premise deployment with zero external dependencies.

How are KSERC regulations applied?

Hardcoded rules: Electricity duty = ■0, CSR/RPO = ■0, O&M; capped per Reg 12(3), 2/3 revenue efficiency share, AT&C; loss benchmarks. When regulations change, we update the rules.

What if PDF extraction fails?

Fallback chain: Claude → Gemini → Groq → GitHub Models → OpenRouter. If all fail, system alerts officer. Manual data entry option available with 'manually verified' marking.

Can officers override AI decisions?

Yes. Simple 'Modify Value' click. New amount with reasoning required. Timestamped, attributed to officer, reversible, full audit trail preserved.

How different from spreadsheets?

Spreadsheets need manual copy-paste. Our tool auto-reads PDFs, understands context, provides reasoning, maintains audit trail, generates documents instantly, and scales to all licensees simultaneously.

Built by RIET for KSERC | Internship Project (Work Order KSERC/CSO/03-05)
GitHub: github.com/prak05/kserc-truing-up